Probabilistic Kernel Optimization for Robust State Estimation

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***Abstract*—Robust state estimation is a fundamental research topic in robotics. Existing approaches like robust kernels combined with iteratively re-weighted least squares (IRLS) often require heuristic parameter selection and extensive fine-tuning. In this manuscript, we propose a novel method that optimizes kernels while preserving the advantages of existing techniques. By introducing a probabilistic interpretation of weights and residuals, our approach enables automatic parameter selection. Applied to iterative closest point (ICP) and bundle adjustment (BA), experimental results demonstrate improved convergence and robustness compared to traditional methods, eliminating the need for time-consuming parameter tuning and offering a practical solution for robust state estimation.**

***Index Terms*—Robust state estimation, SLAM.**

I. Introduction

STATE estimation is one of the fundamental research topics in robotics because of its essential role in navigation for autonomous robots [1]. It is primarily performed through data association between noisy sensor measurements and the non-linear least squares (NLS) method [2]. Since data association in the presence of noise inevitably includes a significant number of outliers, and the NLS method is highly sensitive to them [3], various approaches have been proposed for robust state estimation [4].

A common approach assumes that large residuals are caused by outliers and applies robust kernels to mitigate their impact [4]. These kernels, combined with *iteratively re-weighted least squares* (IRLS) [5], contribute to minimizing the influence of outliers by reducing their weights. The assumption that large residuals are caused by outliers is valid when the initial state is known to a certain degree of accuracy [6]. In such cases, widely used robust kernels such as the *Pseudo-Huber, Cauchy, Welsch, and Geman-McClure* functions are applied [7]. Depending on the situation, these functions are heuristically fine-tuned to appropriately adjust the kernel, or the *median absolute deviation* (MAD) of the residuals is used [8].

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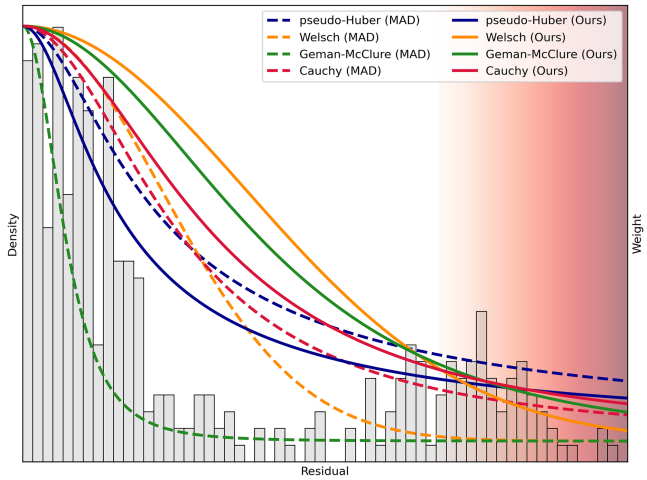


Fig. 1. Histogram (gray) of the residual distribution in the initial state of the *buddha* dataset (Section V-A), together with the weights determined by the MAD-based approach and our proposed method. The white area indicates the inlier region, while colors gradually shifting toward red signify the outlier region. Notably, the MAD-based approach exhibits considerable variation in the weight distribution depending on the choice of kernel, whereas our proposed method maintains a consistent boundary between inliers and outliers.

In real-world scenarios, however, state estimation often fails to predict initial values accurately [9]. In such cases, large residuals may stem from data containing significant patterns rather than outliers, but they may be ignored during the optimization process. Moreover, this approach leads to *non-convex optimization* problems [10]. Non-convex optimization may not guarantee the global minimum and can suffer from slow convergence, especially when the initial value estimation is uncertain [11]. *Graduated non-convexity* (GNC) is an efficient alternative for relaxing non-convex optimization problems. However, it also relies on heuristic decisions for parameters, making fine-tuning essential for its application across various problems [12].

The combination of IRLS and GNC with robust kernels is widely used to improve the robustness of state estimation [12]. However, this approach increases the number of parameters that need to be selected, and as mentioned earlier, it often relies on heuristic approaches. This leads to a time-consuming fine-tuning process, and even after the parameters are determined, their applicability may remain limited. Therefore, there is still room for improvement in achieving practical robust state estimation for real-world applications.

In this manuscript, we propose a novel approach to improve these issues. Our method focuses on adaptively determining

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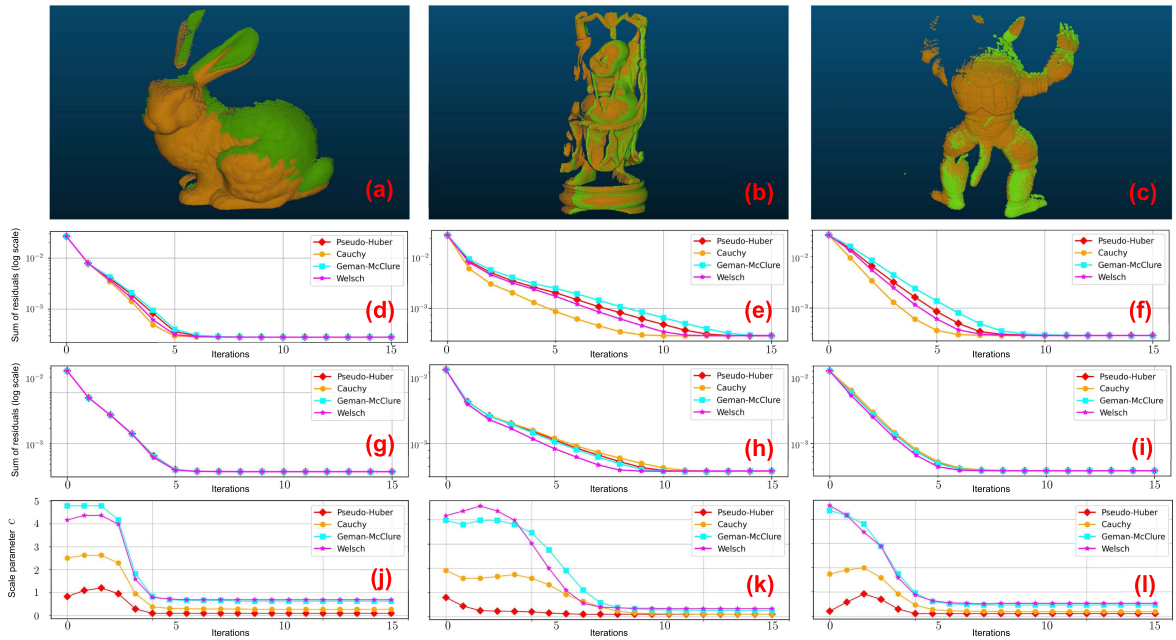


Fig. 2. Registration results and performance evaluation on the *bunny*, *buddha*, and *armadillo* datasets from the Stanford registration dataset. Subfigures (a)– (c) represent the alignment results, with orange point clouds as reference data and green point clouds as source data. Subfigures (d)– (f) and (g)–(i) show the sum of residuals per iteration using MAD-based approach and our method, respectively. Subfigures (j)– (l) illustrate the decreasing trend of scale parameters in the proposed method.

the necessary parameters while preserving the core concepts of existing approaches. We introduce a new probabilistic interpretation of the weights derived from robust kernels and the residuals generated by data association, enabling the automatic selection of optimal parameters without human intervention. We demonstrate the effectiveness of the proposed method by applying it to ICP and BA, two representative state estimation problems in robotics.

II. Related Work

Robust kernels [3], [13], [13] are widely adopted in non-linear least squares to diminish the impact of outliers by assigning high weights to small residuals and down-weighting large ones. Their *scale parameter* demarcates the boundary between convex and non-convex regions [7], necessitating careful tuning to prevent misclassifying valid measurements as outliers [14]. These kernels have proven effective across diverse applications, from LiDAR [15], [16], [17], [18] to vision-based state estimation [19], [20], [21].

When implemented within an IRLS framework, users must select both the kernel type and a suitable scale parameter [8], [22]. Improper selection of the scale parameter can cause valid data to be down-weighted [14], especially when the uncertainty in the initial value estimates is large [12].

Although kernel selection often hinges on sensor characteristics and trial-and-error [23], general approaches such as Barron's adaptive loss [24] and the truncated robust kernel [23] offer more flexibility by blending multiple loss functions. Yet these approaches remain sensitive to parameter tuning and the residual scale and selection of scale parameter [23].

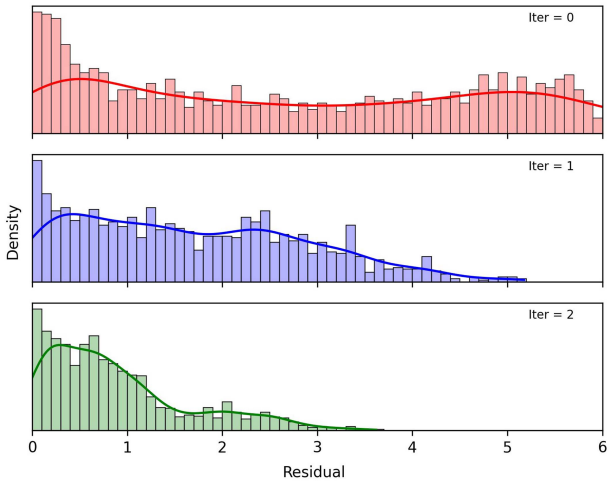


Fig. 3. Residual histograms and density estimates across iterations for the *bunny* dataset, as described in Section V-A.

Recent studies have focused on adaptively determining the kernel for robust state estimation. However, we argue that determining the scale parameter, which serves as the criterion for distinguishing outliers from inliers, is more crucial than the choice of kernel.

A classical approach to determining the scale parameter is to use a method based on the median absolute deviation (MAD) [8]. This is especially effective for simple, symmetric distributions [25], but its robustness may suffer under asymmetric or multi-modal residuals [26], particularly with poor initial guesses.

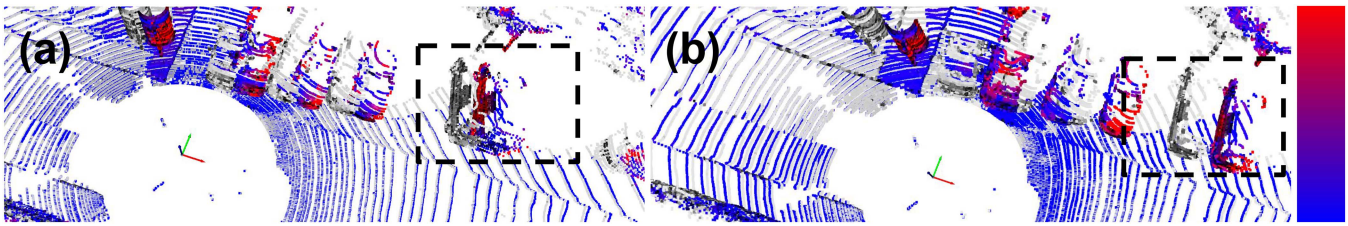


Fig. 4. Initial stage residuals between frames 45 and 46 (a), and between frames 45 and 47 (b) in sequence 5 of the KITTI odometry dataset, showing forward movements of about 1.06 m and 2.01 m, respectively. The source-frame points are colored by residual magnitude, with smaller residuals in blue and larger ones in red. The black dotted lines help visualize the ground-truth forward distance.

Because robust kernels inherently introduce non-convex cost functions [10], minimizing them poses a risk of converging to local minima [11]. Graduated non-convexity (GNC) addresses this issue by starting with an L_2-like convex approximation and gradually reverting to the original robust kernel. Yang et al. [12] demonstrate that GNC can effectively handle large initial errors. However, it also depends on additional transition parameters that often require fine-tuning [12].

III. Background

In this section, we revisit robust non-linear least squares and graduated non-convexity (GNC) for non-convex optimization, elaborating on the core principles of these methods and their relevance to parameter optimization. This provides the necessary theoretical foundation for the method introduced in this paper.

A. Robust Non-Linear Least Squares

State estimation in robotics is frequently framed as a non-linear least squares (NLS) problem, where the objective is to minimize the sum of squared residuals [2]. This formulation can be written as:

$$\theta^* = \arg\min_{\theta} \frac{1}{2} \sum_{i=1}^{N} w_i ||r_i(\theta)||^2 \tag{1}$$

where w_i is the weight applied to the i-th residual $r_i(\theta)$. Residuals result from the data associations between sensor measurements. Due to sensor noise and incorrect data associations, a considerable number of outliers are inevitably included. These outliers negatively impact the optimization process as they incur higher penalties in NLS optimization [3].

To mitigate the adverse effects of outliers in optimization, robust kernels such as Pseudo-Huber [13], Cauchy [4], Welsch [5], and Geman-McClure [27] are commonly employed. These kernels transition from convex to non-convex behavior depending on a scale parameter, which results in reduced gradient magnitudes for large residuals. In the iteratively re-weighted least squares (IRLS), large residuals, typically regarded as outliers, are assigned smaller weights, thereby diminishing their influence on the optimization process [5].

The formulation for robust optimization is expressed as follows:

$$\theta^* = \arg\min_{\theta} \sum_{i=1}^{N} \rho(r_i(\theta)) \tag{2}$$

where $\rho(r)$ is a robust kernel. This robust optimization approach can be incorporated into IRLS by ensuring gradient consistency between (1) and (2) [5]. The solution can then be derived using standard optimization techniques, such as Gauss-Newton or Levenberg-Marquardt [11]. In this framework, the weights are computed as follows:

$$w_i = \frac{1}{r_i(\theta)} \rho'(r_i(\theta)) \tag{3}$$

The conventional method for determining the scale parameter involves either fine-tuning for the specific problem or calculating the median absolute deviation (MAD) of the residuals at each iteration, followed by scaling with a kernel-specific constant [8]. This MAD-based approach performs effectively without the need for fine-tuning when the residual distribution is symmetric and unimodal [25]. However, when the initial estimate is inaccurate, the residual distribution can become multi-modal and asymmetric, exposing the limitations of this method and the need for potential improvements [26].

B. Graduated Non-Convexity (GNC)

Graduated Non-Convexity (GNC) is a widely adopted technique for optimizing general non-convex cost functions $\rho(\cdot)$, and has been applied in diverse fields such as computer vision and machine learning. The key idea behind GNC is to introduce a *control parameter* to a robust kernel, allowing the optimization problem to transition from a convex to a progressively non-convex form. The method starts by optimizing a convex surrogate function, using the solution from the previous step as the initialization for the next, and finally solving the original non-convex problem [12].

One of the most commonly used robust kernels in GNC is the Geman–McClure function. We denote it by $\rho(r)$ with the scale parameter c and the residual r. In GNC, a control parameter μ is introduced to define the surrogate cost function $\rho_\mu(r)$. Both can be expressed as:

$$\rho(r) = \frac{c^2 r^2}{c^2 + r^2} \quad \text{and} \quad \rho_\mu(r) = \frac{\mu c^2 r^2}{\mu c^2 + r^2}. \tag{4}$$

This surrogate function $\rho_\mu(r)$ exhibits two key properties. First, when μ is sufficiently large, $\rho_\mu(r)$ becomes convex and converges to a *quadratic form* [12]. Second, when $\mu = 1$, $\rho_\mu(r)$ recovers the original non-convex cost function $\rho(r)$.

This approach has demonstrated its effectiveness in several state estimation problems; however, it introduces the additional

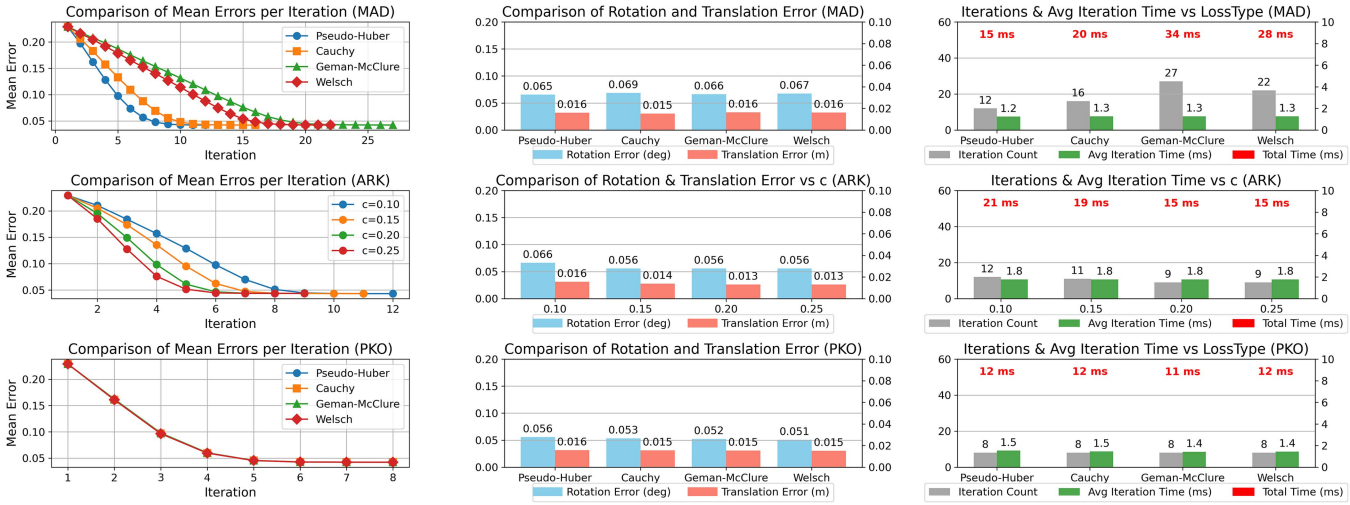


Fig. 5. From top to bottom, these results were obtained by applying the MAD-based method, the adaptive robust kernel (ARK) [23], and the probabilistic kernel optimization (PKO) proposed in this paper to dataset A in Fig. 4.

challenge of parameter selection [12]. These parameters are typically determined heuristically, which necessitates fine-tuning in practical applications. GNC generally operates by keeping the c value fixed and progressively varying μ throughout the optimization [12]. However, as can be inferred from (4)–(5), varying μ and varying c produce the same shape transformation in this function. This implies that by initializing the scale parameter c with a large value and progressively reducing it throughout the IRLS iterations, the kernel can be updated in a manner consistent with the GNC mechanism, thereby eliminating the need for additional parameters.

IV. Probabilistic Kernel Optimization

A. Residual Distribution in IRLS

In Iteratively Reweighted Least Squares (IRLS), residual weights are recalculated at each iteration, typically based on kernel forms adjusted using heuristic methods or values derived from the median absolute deviation (MAD) [5]. MAD effectively determines the scale parameter when the residual distribution is symmetric and uni-modal, resembling a normal distribution [25]. However, when uncertainty exists in the initial state, as in LiDAR ego-motion estimation where consecutive point clouds are registered over time [15], the residual distribution may deviate from this ideal assumption, becoming irregular and multi-modal.

Fig. 3 shows the residual histograms and kernel density estimation (KDE) distributions for the *bunny* dataset from Section V-B. This illustrates that, starting from an irregular and multi-modal residual distribution at the initial stage, the distribution gradually converges toward zero as the iterations progress.

This analysis highlights the importance of computing weights that accurately represent the residual distribution at each IRLS iteration for robust state estimation. Early iterations may feature non-symmetric, multi-modal residuals, necessitating adjustments to the robust kernel to correctly identify valid residuals based on their density, regardless of magnitude. This approach offers two advantages: (i) it promotes optimization towards a global solution even with uncertain initial values, and (ii) as iterations progress, the residual distribution converges to a zero-mean normal distribution, allowing for an adaptive reduction of the scale parameter. This process aligns with the concept of graduated non-convexity (GNC) discussed in Section III and can be implemented without additional parameters.

B. Probabilistic Interpretation of Weight and Residual

Building upon the foundational concept of the IRLS framework—which aims to assign higher weights to inliers and lower weights to outliers to reduce the influence of outliers—we propose a probabilistic interpretation of the weighting scheme. Specifically, we interpret the weight w_i as being proportional to the probability that the residual r_i is an inlier, given a scale parameter c:

$$w_i \propto P_{\text{model}}(\text{inlier} \mid r_i, c). \tag{5}$$

This expression is considered the *model-based probability* because it is derived from the weighting function w_i used within the IRLS algorithm, which is an inherent component of the model's formulation [25]. The weighting function depends on the choice of the robust kernel and the scale parameter c, both defined by the model to handle residuals effectively [26]. Thus, $P_{\text{model}}(\text{inlier} \mid r_i, c)$ reflects how the model assesses the likelihood of each residual being an inlier based on its internal mechanisms.

To formalize this interpretation, we normalize the weights to ensure they represent valid probability distributions:

$$P_{\text{model}}(\text{inlier} \mid r_i, c) = \frac{w_i}{Z_w}, \tag{6}$$

where $Z_w = \int_0^\infty w_i \, dr_i$ is the normalization constant ensuring that the probabilities sum to one over all possible residuals. Note that we consider only absolute value of residuals in this formulation, as indicated by the integration limits.

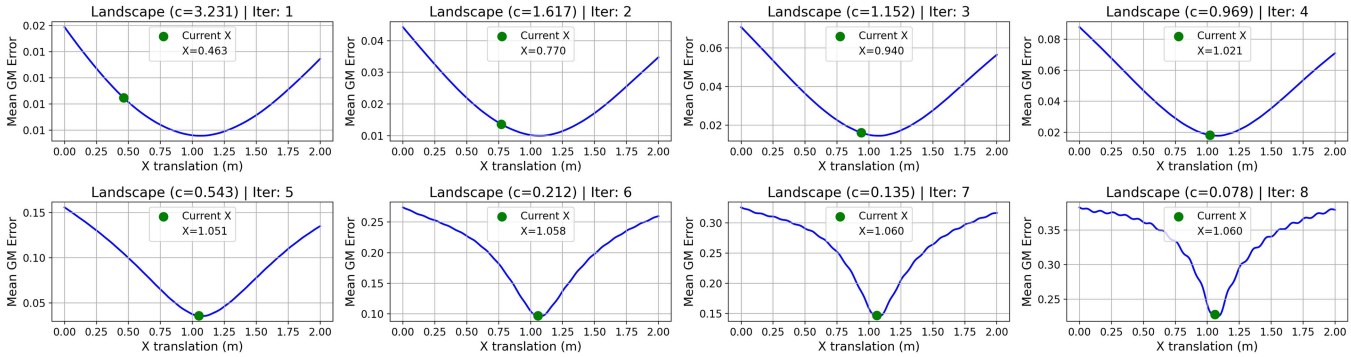


Fig. 6. This illustrates the eight-step optimization process using our proposed method with the Geman–McClure kernel in Fig. 4(a).

Similarly, we consider the residual distribution from a data-driven probabilistic perspective. As illustrated in Fig. 3, residuals with higher density values are more likely to be inliers containing significant patterns within the data.

This idea is based on the statistical learning theory [28]. The core principles of this theory can be summarized as Empirical Risk Minimization (ERM) and Structural Risk Minimization (SRM).

Given a dataset $\{(\mathbf{x}_i, y_i)\}_{i=1}^n$, the empirical risk is defined as:

$$R_{\text{emp}}(f) = \frac{1}{n}\sum_{i=1}^{n} L\big(f(\mathbf{x}_i), y_i\big), \tag{7}$$

where $L(\cdot,\cdot)$ is a loss function, and f is the model function. The Vapnik–Chervonenkis theory states that the difference between the true risk $R(f)$ and the empirical risk $R_{\text{emp}}(f)$ is probabilistically bounded as follows:

$$P\{|R(f) - R_{\text{emp}}(f)| \le \epsilon\} \ge 1 - 2e^{-2n\epsilon^2}. \tag{8}$$

As n increases, $R_{\text{emp}}(f)$ converges to $R(f)$. Additionally, in the context of statistical learning theory, the following holds in a form analogous to the Law of Large Numbers:

$$P\Big(\sup_{f\in\mathcal{F}}\big|R_{\text{emp}}(f) - R(f)\big| > \epsilon\Big) \le 2\mathcal{N}(\mathcal{F}, n)\, e^{-\frac{n\epsilon^2}{8}}, \tag{9}$$

where $\mathcal{N}(\mathcal{F}, n)$ denotes the growth function of the function class $\mathcal{F}$.

Intuitively, the fundamental idea of statistical learning theory is that patterns frequently observed in the data are more likely to reflect the true data generating process.

This theoretical result supports the mathematical relationship between the density of the residual distribution and measurement reliability [28].

Based on this, we assume that the probability density function $f(r_i)$ representing the distribution of residuals corresponds to the probability that a residual r_i is an inlier:

$$P_{\text{data}}(\text{inlier} \mid r_i) = f(r_i). \tag{10}$$

This expression is considered the *data-based probability* because it is estimated directly from the observed data without relying on model assumptions. Specifically, $f(r_i)$ is obtained by modeling the empirical distribution of the residuals r_i collected from the data. It captures the inherent patterns present in the data, reflecting the actual likelihood of each residual being an inlier based on observed evidence.

Algorithm 1: Probabilsitic Kernel Optimization.

Require: Residuals $\{r_i\}_{i=1}^M$, robust kernel $\kappa(r; c)$
Ensure: Optimized scale parameter c^*
// **Step 1: Fit a GMM to obtain $P_{\text{data}}(\text{inlier} \mid r)$**
$\{\pi_j, \mu_j, \sigma_j^2\}_{j=1}^N \leftarrow \text{EMFitGMM}\big(\{r_i\}_{i=1}^M\big)$
$P_{\text{data}}(\text{inlier} \mid r) = \sum_{j=1}^N \pi_j \mathcal{N}(r \mid \mu_j, \sigma_j^2)$
// **Step 2: Define P_{model} using the robust kernel**
$Z_w = \int_0^\infty w(r; c)$
$P_{\text{model}}(\text{inlier} \mid r, c) = \frac{w}{Z_w}$
// **Step 3: Search for the optimal c that minimizes JS divergence**
candidateSet $\leftarrow$ GenerateGridOrRange$(c_{\min}, c_{\max})$
$c^* \leftarrow \underset{c\,\in\,\text{candidateSet}}{\arg\min}\; JS\big(P_{\text{data}}(\cdot)||P_{\text{model}}(\cdot \mid c)\big)$
return c^*

To approximate $f(r_i)$, we use the expectation-maximization (EM) algorithm [29] to fit a Gaussian mixture model (GMM) to the residuals. For computational efficiency, rather than using all residuals, we adopt a sampling-based approach. In our method, the probability that r_i is an inlier is defined as:

$$P_{\text{data}}(\text{inlier} \mid r_i) = \sum_j \pi_j \mathcal{N}(r_i \mid \mu_j, \sigma_j^2). \tag{11}$$

C. Probabilistic Kernel Optimization

In this paper, we propose a method to adaptively determine the scale parameter c of a robust kernel at each iteration of the IRLS algorithm, based on proposed probabilistic interpretations. This approach minimizes the difference between the model-based probability distribution $P_{\text{model}}(\text{inlier} \mid r_i, c)$ and the data-based probability distribution $P_{\text{data}}(\text{inlier} \mid r_i)$. By aligning these two distributions, the model can better reflect the actual data distribution.

Typically, the Kullback-Leibler (KL) divergence measures the difference between two probability distributions, but it can be asymmetric and overly sensitive, causing overfitting [30], [31],

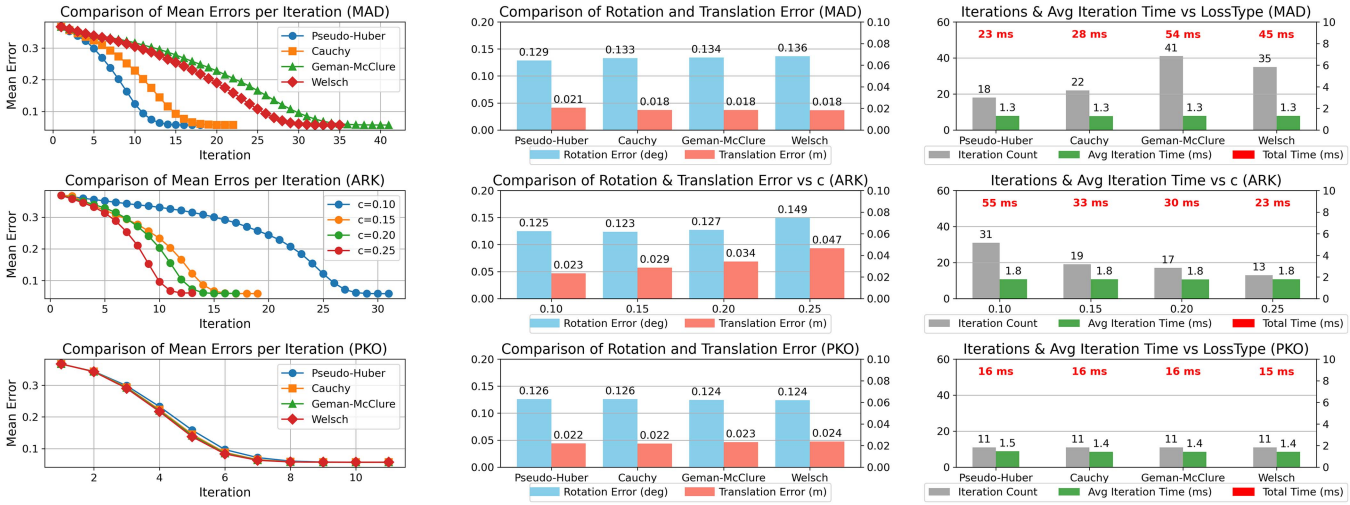


Fig. 7. From top to bottom, these results were obtained by applying the MAD-based method, the adaptive robust kernel (ARK) [23], and the probabilistic kernel optimization (PKO) proposed in this paper to dataset B in Fig. 4.

[32], [33]. To mitigate this, we employ the Jensen-Shannon (JS) divergence, a symmetric and always finite variant of KL. By averaging the two distributions, the JS divergence provides a more stable measure and reduces over-sensitivity to specific data patterns [34], [35].

Therefore, in the process of determining the scale parameter c of the robust kernel, we adopt a method that minimizes the JS divergence between $P_{\text{data}}(\text{inlier} \mid r_i)$ and $P_{\text{model}}(\text{inlier} \mid r_i, c)$. This is expressed mathematically as:

$$c = \arg\min_{c} JS(P_{\text{data}}(\text{inlier} \mid r_i) \parallel P_{\text{model}}(\text{inlier} \mid r_i, c)) \quad (12)$$

By determining the scale parameter c using this approach, the model can more accurately reflect the actual data distribution. Minimizing the JS divergence assists the model in balancing various characteristics of the data, and experiments have confirmed that this approach leads to improved model performance. A schematic algorithm describing this process is presented in Algorithm 1.

V. Experimental Result

In this manuscript, we conduct experiments to demonstrate that the proposed method can achieve robust state estimation without the need for hyper-parameter tuning. In the first experiment, we utilize the Stanford registration dataset [36] and a subset of LiDAR point clouds from the KITTI odometry dataset to evaluate the performance of our method in the context of ICP for partially overlapping point clouds. The second experiment applies the proposed algorithm to LiDAR odometry using ICP and visual odometry using Bundle Adjustment (BA) on the KITTI odometry dataset [37], demonstrating its practical applicability in real-world scenarios.

A. Point Cloud Registration

We use the *bunny*, *buddha*, and *armadillo* data from the Stanford registration dataset [36] to align two point clouds by minimizing the point-to-plane distance. Fig. 2 shows the alignment results and the sum of residuals over iterations for different kernels, comparing MAD-based approach and our proposed method. The MAD-based approach exhibits larger variations depending on the kernel's outlier-handling strategy, whereas our method consistently reduces variability in convergence. As shown in Fig. 1, this is because our proposed method can assign appropriate and consistent weights to frequently occurring residuals, regardless of the chosen kernel, even under complex residual distributions. Our method also decreases the scale parameter as iterations progress, aligning with the idea of increasing non-convexity in GNC, yet requires no additional parameters.

We also conducted experiments on two samples from the KITTI odometry dataset, as illustrated in Fig. 4. Notably, dataset (b) involves a more uncertain initial state estimate (e.g., larger pose discrepancy between consecutive frames), making the registration task more challenging. As shown in Figs. 5 and 7, our method (PKO) not only outperforms the MAD-based approach in terms of accuracy, convergence, and computational efficiency, but also exhibits minimal variability across different kernel choices, even under higher uncertainty in the initial state estimate. We further applied the adaptive robust kernel (ARK) proposed in [23] to this experiment. While ARK offers the advantage of adaptively determining the kernel shape, it exhibits greater variations in state estimation accuracy and convergence for relatively uncertain initial states. For this reason, in the subsequent full-sequence evaluations on the KITTI odometry dataset, we compare only our method with the MAD-based method. This decision stems from the fact that the results with ARK can vary with different scale parameter settings, making it difficult to perform quantitative comparison.

We illustrate the eight-step solution convergence in Fig. 6 using the Geman–McClure kernel in (a), demonstrating that our method can incorporate the GNC concept without additional parameters or heuristic scheduling. Because the motion is

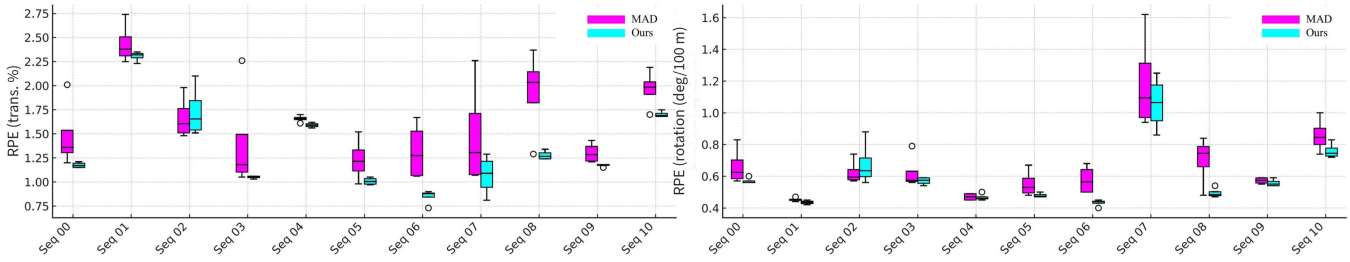


Fig. 8. Boxplots of RPE across sequences 00–10 using LiDAR odometry.

TABLE I
RELATIVE TRANSLATION ERROR (%) AND ROTATION ERROR (DEGREE/100 M) FOR EACH SEQUENCE USING LIDAR ODOMETRY

Sequence (Num. of frames)	00 (4541)	01 (1011)	02 (801)	03 (4661)	04 (271)	05 (2761)	06 (1101)	07 (1101)	08 (4071)	09 (1591)	10 (1201)	Avg.
Geman-McClure (MAD)	2.01/0.83	2.74/0.45	1.98/0.74	2.26/0.79	1.66/0.49	1.52/0.67	1.67/0.68	2.26/1.62	2.37/0.84	1.43/0.59	2.19/0.82	2.01/0.77
Geman-McClure (Ours)	1.21/0.56	2.31/0.44	1.76/0.66	1.06/0.59	1.62/0.46	1.03/0.50	0.88/0.44	1.29/1.25	1.34/0.54	1.18/0.56	1.68/0.76	**1.40/0.61**
Welsch (MAD)	1.34/0.59	2.43/0.45	1.52/0.58	1.12/0.57	1.70/0.49	1.16/0.50	1.06/0.50	1.07/0.98	2.07/0.77	1.22/0.56	1.98/0.74	1.52/0.61
Welsch (Ours)	1.15/0.60	2.23/0.42	2.10/0.88	1.03/0.56	1.58/0.50	0.97/0.48	0.73/0.40	0.81/0.86	1.24/0.49	1.15/0.59	1.75/0.72	**1.34/0.59**
Cauchy (MAD)	1.20/0.57	2.33/0.47	1.48/0.57	1.05/0.58	1.61/0.45	0.98/0.48	1.07/0.50	1.08/0.94	1.29/0.48	1.21/0.55	1.70/1.00	1.36/0.60
Cauchy (Ours)	1.15/0.56	2.35/0.45	1.55/0.61	1.05/0.59	1.60/0.45	0.98/0.47	0.88/0.44	1.19/1.15	1.24/0.47	1.18/0.54	1.70/0.73	**1.35/0.59**
PseudoHuber (MAD)	1.38/0.66	2.25/0.44	1.69/0.61	1.24/0.56	1.66/0.45	1.27/0.56	1.48/0.63	1.53/1.21	2.00/0.72	1.35/0.59	1.99/0.87	1.62/0.66
PseudoHuber (Ours)	1.19/0.56	2.33/0.43	1.51/0.56	1.06/0.54	1.56/0.46	1.05/0.47	0.90/0.45	0.99/0.98	1.29/0.48	1.18/0.54	1.68/0.83	**1.34/0.57**

TABLE II
RELATIVE TRANSLATION ERROR (%) AND ROTATION ERROR (DEGREE/100 M) FOR EACH SEQUENCE USING VISUAL ODOMETRY

Sequence (Num. of frames)	00 (4541)	01 (1011)	02 (801)	03 (4661)	04 (271)	05 (2761)	06 (1101)	07 (1101)	08 (4071)	09 (1591)	10 (1201)	Avg.
Geman-McClure (MAD)	1.04/0.35	1.78/0.38	1.04/0.32	1.24/0.26	1.49/0.37	0.95/0.31	1.05/0.31	1.17/0.60	1.19/0.35	1.58/0.30	0.63/0.32	1.20/0.35
Geman-McClure (Ours)	1.03/0.35	1.55/0.29	1.02/0.32	1.23/0.27	1.34/0.31	0.84/0.29	0.94/0.31	0.97/0.55	1.21/0.36	1.40/0.30	0.72/0.30	**1.11/0.33**
Welsch (MAD)	1.14/0.43	1.82/0.39	1.20/0.39	1.31/0.48	1.53/0.42	0.89/0.30	1.07/0.39	0.93/0.55	1.32/0.39	1.28/0.23	0.69/0.32	1.20/0.39
Welsch (Ours)	1.03/0.37	1.43/0.35	1.06/0.36	1.13/0.21	1.40/0.33	0.83/0.26	0.90/0.34	0.92/0.52	1.22/0.36	1.37/0.26	0.77/0.33	**1.10/0.34**
Cauchy (MAD)	0.90/0.33	1.41/0.37	0.91/0.30	1.08/0.30	1.41/0.30	0.89/0.30	0.90/0.37	1.44/0.49	1.26/0.38	1.31/0.26	0.87/0.55	1.13/0.36
Cauchy (Ours)	0.86/0.30	1.21/0.33	0.96/0.31	1.07/0.24	1.38/0.33	0.92/0.31	0.85/0.32	0.75/0.42	1.18/0.34	1.29/0.26	0.74/0.30	**1.02/0.31**
PseudoHuber (MAD)	0.97/0.33	1.52/0.39	0.93/0.30	1.22/0.33	1.39/0.32	0.89/0.30	0.95/0.29	0.96/0.48	1.27/0.39	1.33/0.26	0.76/0.33	1.11/0.34
PseudoHuber (Ours)	0.94/0.33	1.31/0.31	1.00/0.31	1.12/0.26	1.40/0.34	0.85/0.29	0.90/0.34	0.93/0.51	1.23/0.36	1.34/0.28	0.84/0.35	**1.08/0.33**

predominantly in the x-direction, we plot the error along x. Under our method, the scale parameter starts large in the initial iterations and gradually decreases. As a result, the optimization begins in a convex regime and progressively becomes non-convex, thereby improving the chance of converging to the global optimum and promoting rapid convergence.

This result is consistent with the following claims we make in our paper:

- It is more important to appropriately assign the scale parameter than to focus on the choice of kernel, in order to improve the performance of robust state estimation.
- Our proposed method can apply the idea of GNC to any robust kernel without additional parameters, thereby increasing the chance of finding the global minima.

B. KITTI Odometry Dataset

To validate the applicability of our proposed algorithm on real-world data, we applied it to sequences 0–10 of the KITTI odometry dataset [37], which includes motion-compensated LiDAR scans, stereo camera images, and calibration data. As detailed in Section V-A, we estimated LiDAR and visual odometry using ICP for point-to-plane distance minimization and Bundle Adjustment (BA), respectively. For LiDAR odometry, we followed the feature extraction, registration, and mapping procedures outlined in LOAM [15]. In visual odometry, we adopted a structure similar to that of ORB-SLAM [21].

Tables I and II present a comparative analysis of the MAD-based method and our proposed method for LiDAR and visual odometry on the KITTI sequences. Odometry accuracy was evaluated using the relative pose error, as recommended by the KITTI odometry benchmark [37]. Figs. 8 and 9 represent these errors for each sequence as boxplots, illustrating the results obtained with different kernels. All other parameters unrelated to the kernel, such as the voxel downsampling intervals in ICP, were kept consistent across all sequences for both methods. As demonstrated by these tables and figures, our proposed algorithm not only enhances overall performance without requiring extensive parameter tuning but also reduces the variation in accuracy across different kernels.

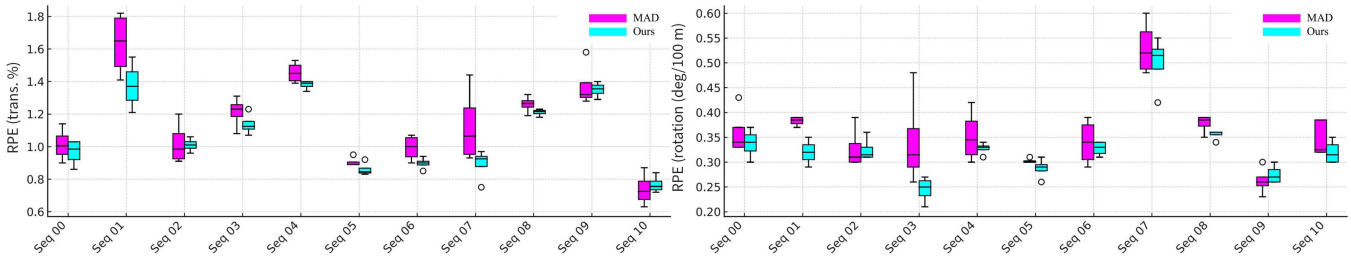


Fig. 9. Boxplots of RPE across sequences 00–10 using visual odometry.

VI. Conclusion

In this paper, we proposed a probabilistic kernel optimization technique for practical and robust state estimation in real-world scenarios without the need for traditional parameter fine-tuning processes. By introducing a novel probabilistic interpretation of residuals and weights, we developed a data-driven method to adaptively optimize the kernel function. Our experiments provided detailed insights into the mechanism of our approach and demonstrated its effectiveness compared to existing methods on real-world datasets. This work contributes to practical applications by alleviating the need for time-consuming parameter fine-tuning in robust state estimation.

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